Finalterm preparations 2024

STA 301

Virtual University

2024

Quiz 4 solution

Module 31 to 35 50 MCQs

Time Left sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:13 AM, 04 January 2025**

Question # 1 of 10 (Start time: 01:13:59 AM, 04 January 2025) **Total Marks: 1**

Suppose we have a population of size N = 5 i.e. 0,11,20,23,45 and if we draw all possible samples of size n = 2 with replacement then total possible samples will be:

Select the correct option

- ○ 50
- ○ 10
- ○ 35
- ◉ 25

Click To Save Answer & Move to Next Question

BC240221728: AHSAN MOBEEN

Time Left 65 sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:13 AM, 04 January 2025

Question # 2 of 10 (Start time: 01:14:28 AM, 04 January 2025)

Total Marks: 1

In general sample variance is a/an .....estimator?

Select the correct option

- Consistent
- none of these
- Unbiased
- Biased

Click to Save Answer & Move to Next Question

BC240221728: AHSAN MOBEEN

Time Left 66 sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:13 AM 14 January 2025

Question # 3 of 10 (Start time: 01:14:55 AM, 04 January 2025)

Total Marks: 1

Assume a large distribution in which σ = 6. What is the standard error of the mean based on samples for which n = 10?

Select the correct option

Reload Math Equations

- 167
- 1.9
- 3
- 15

Click to Save Answer & Move to Next Question

BC240221728: AHSAN MOBEEN

Time Left 62 sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:13 AM, 04 January 2025

Question # 4 of 10 (Start time: 01:15:21 AM, 04 January 2025)

Total Marks: 1

The relative efficiency of T1 compared to T2 where:

Select the correct option

- T1 is unbiased estimator
- T2 is unbiased estimator
- both T1 and T2 are unbiased estimators
- both T1 and T2 are biased estimators

Saving...

sec(s)

STA301 - Statistics and Probability (Quiz 4) Quiz Start Time: 01:13 AM, 04 January 2025

Question # 5 of 10 (Start time: 01:15:50 AM, 04 January 2025) T al Marks: 1

If μ1 = 225 and μ2 = 186, the difference of population means will be:

Select the correct option

- (●) 39
- ( ) 57
- ( ) 29
- ( ) 25

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:13 AM, 04 January 2025**

Question # 6 of 10 (Start time: 01:16:16 AM, 04 January 2025) **T al Marks:**

Which of the following statements is correct?

Select the correct option

- ○ All of the above statements are correct
- ○ A point estimate is an estimate of the range of a population parameter
- ○ A point estimate is an unbiased estimator if its standard deviation is the same as the actual value of the population standard deviation
- ◉ A point estimate is a single value estimate of the value of a population parameter

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:13 AM, 04 January 2025

Question # 7 of 10 (Start time: 01:16:52 AM, 04 January 2025)

T al Marks: 1

Ideally, the width of confidence interval should be

Select the correct option

- 98
- 0
- 1
- 100

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:13 AM, 04 January 2025**

Question # 8 of 10 (Start time: 01:17:15 AM, 04 January 2025) **T al Marks: 1**

"A point estimate plus/minus a few times the standard error of that estimate". This statement represents:

Select the correct option

- ○ Acceptance region
- ○ Critical region
- ◉ Confidence interval
- ○ Critical value

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:13 AM, 04 January 2025**

Question # 9 of 10 (Start time: 01:17:42 AM, 04 January 2025) **T al Marks: 1**

Total no. of possible samples of size 3 (with replacement) from the population of size 6, will be:

Select the correct option

- ○ 325
- ○ 196
- ○ 256
- ◉ 216

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:13 AM, 04 January 2025**

Question # 10 of 10 (Start time: 01:18:06 AM, 04 January 2025) **T al Marks:**

In sampling from a large population with $\sigma = 35$, the standard error of the mean is found to be 4. The size of the sample used is:

Select the correct option

- 15
- 56
- 25
- 77

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:26 AM, 04 January 2025**

Question # 1 of 10 (Start time: 01:26:01 AM, 04 January 2025) **T al Marks: 1**

If the two populations are non-normal and both the sample sizes are large, then the sampling distribution of the difference of means will be:

Select the correct option

- ○ t-distribution
- ○ F-distribution
- ◉ Normal distribution
- ○ Chi-square distribution

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:26 AM, 04 January 2025**

Question # 2 of 10 (Start time: 01:26:31 AM, 04 January 2025) **T al Marks:**

A statistic whose standard error decreases with an increase in the sample size will be .....

Select the correct option

- None
- Consistant
- Unbiased
- Efficient

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:26 AM, 04 January 2025**

Question # 3 of 10 (Start time: 01:26:57 AM, 04 January 2025) **Total Marks:** 1

The difference between the value of the statistic and the value of the parameter is called _____.

Select the correct option

- ○ Random Error
- ○ Standard Error
- ◉ Sampling Error
- ○ Biased Error

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:26 AM, 04 January 2025**

Question # 4 of 10 (Start time: 01:27:26 AM, 04 January 2025) **T al Marks: 1**

When sampling is from an infinite population, the mean and standard error of the sampling distribution of p^ are _____ and_____, respectively:

Select the correct option

- p, n(1 - p)p
- p, sqrt [p(1 - p)/n]
- p^, p(1 - p) / n
- p, p(1 - p)

Click to Save Answer & Move to Next Question

BC230409381: ASIF MEHMOOD

Time Left 59 sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:26 AM 4 January 2025

Question # 5 of 10 (Start time: 01:27:54 AM, 04 January 2025)

Total Marks: 1

In sampling from a large population with sigma = 35, the standard error of the mean is found to be 4 text. The size of the sample used is:

Select the correct option

Reload Math Equations

- 15
- 25
- 56
- 77

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:26 AM, 04 January 2025**

Question # 6 of 10 (Start time: 01:28:28 AM, 04 January 2025) **Total Marks:** 1

If μ1 = 225 and μ2 = 186, the difference of population means will be:

Select the correct option

- ○ 29
- ○ 57
- ◉ 39
- ○ 25

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:26 AM, 04 January 2025**

Question # 7 of 10 (Start time: 01:28:54 AM, 04 January 2025) **T al Marks:**

A quantity obtained by applying certain rule or formula is known as

Select the correct option

- ○ Estimator
- ○ Parameter
- ◉ Estimate
- ○ Proportion

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:26 AM, 04 January 2025**

Question # 8 of 10 (Start time: 01:29:17 AM, 04 January 2025) **T al Marks:**

The sampling distribution of the mean becomes approximately normally distributed only when the following conditions is met?

Select the correct option

- ○ The population is normally distributed.
- ○ A single random sample is drawn from the population.
- ◉ The sample size is large.
- ○ The standard deviation of the population is large.

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:26 AM, 04 January 2025**

Question # 9 of 10 (Start time: 01:29:42 AM, 04 January 2025) **T al Marks: 1**

Which of the following statements about confidence intervals is inaccurate?

Select the correct option

- ○ If we keep the sample size fixed, the confidence interval gets wider as we increase the con?dence coefficient
- ◉ If the population standard deviation increases, the confidence interval decreases in width
- ○ If we keep the confidence coefficient fixed, the confidence interval gets narrower as we increase the sample size
- ○ A confidence interval for a mean always contains the sample mean

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:26 AM, 04 January 2025**

Question # 10 of 10 (Start time: 01:30:18 AM, 04 January 2025) **Total Marks:** 1

Method of Moments for estimation of parameters is based on the theory of

Select the correct option

- ○ Non-linear estimation
- ◉ Linear Estimation
- ○ None of the above
- ○ Both Linear and Non-linear

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:32 AM, 04 January 2025**

Question # 1 of 10 (Start time: 01:32:57 AM, 04 January 2025) **Total Marks:** 1

Sampling distributions are:

Select the correct option

- (●) The probability distribution of sample statistic
- ( ) The probability distribution that is assigned to population parameters
- ( ) Refers to the margine of error
- ( ) None of above

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:32 AM, 04 January 2025**

Question # 2 of 10 (Start time: 01:33:26 AM, 04 January 2025) **Total Marks: 1**

"To consider every possible value that the parameter might have, and for each value, compute the probability and THAT value of the parameter for which the probability of a given sample is greatest, is chosen as an estimate." This procedure is known as

Select the correct option

- ○ The method of moments
- ○ The method of fractional moments
- ◉ The method of maximum likelihood
- ○ The method of least square

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:32 AM, 04 January 2025**

Question # 3 of 10 (Start time: 01:33:52 AM, 04 January 2025) **T al Marks: 1**

In sampling from a large population with $\sigma = 25$. The size of the sample used is 10, the value of the standard error will be:

Select the correct option

- (●) 7.9
- ( ) 0
- ( ) 6
- ( ) 2

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) Quiz Start Time: 01:32 AM, 04 January 2025

Question # 4 of 10 (Start time: 01:34:23 AM, 04 January 2025) T al Marks: 1

In sampling without replacement; fpc is used when:

Select the correct option

- n<0.05N
- n=0.05N
- None of the above
- n>0.05N

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:32 AM, 04 January 2025**

Question # 5 of 10 (Start time: 01:34:48 AM, 04 January 2025) **T al Marks: 1**

Consider a large population with a mean of 160 and a standard deviation of 25. A random sample of size 64 is taken from this population. What is the standard deviation of the sample mean?

Select the correct option

- ○ 2.569
- ○ 1.256
- ◉ 3.125
- ○ 4.254

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:32 AM, 04 January 2025**

Question # 6 of 10 (Start time: 01:35:19 AM, 04 January 2025) **T al Marks: 1**

P 0 1/3 2/3 1
f(p) 1/10 4/10 4/10 1/10

The mean of the sampling proportion will be:

Select the correct option

- ○ 1/20
- ◉ 15/30
- ○ 10/20
- ○ 1/15

Click to Save Answer & Move to Next Question

BC230422770: OSAMA KHAN

Time Left 64 sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:32 AM 04 January 2025

Question # 7 of 10 (Start time: 01:35:53 AM, 04 January 2025)

Total Marks: 1

In sampling from a large population with sigma = 20, the standard error of the mean is found to be 2. The size of the sample used is:

Select the correct option

Reload Math Equations

- ○ 10
- ○ 20
- ○ 40
- ◉ 100

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:32 AM, 04 January 2025**

Question # 8 of 10 (Start time: 01:36:21 AM, 04 January 2025) **Total Marks:** 1

Which of the following is not a criterion for a good problem situation?

Select the correct option

- ○ It must be such that its solution will be benefited by group effort.
- ○ It must be appropriate to the students' level of development.
- ○ It must be authentic.
- ◉ It must be clearly structured and defined.

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:32 AM, 04 January 2025**

Question # 9 of 10 (Start time: 01:36:51 AM, 04 January 2025) **T al Marks: 1**

A _____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

Select the correct option

- (●) Confidence interval
- ( ) Probability limit
- ( ) Lower limit
- ( ) Hypothesis

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:32 AM, 04 January 2025**

Question # 10 of 10 (Start time: 01:37:15 AM, 04 January 2025) **T al Marks:**

Statistical Inference can be divided into two main branches:

Select the correct option

- ○ All above
- ○ Interval Estimation and Point Estimation
- ○ MLE and Methods of moments
- ◉ Estimation and Hypothesis testing

Click to Save Answer & Move to Next Question

BC230206575: SIDRA YOUSAF

Time Left 58 sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:40 AM **4 J** **uary 2025**

Question # 1 of 10 (Start time: 01:40:04 AM, 04 January 2025) **Total Marks: 1**

The standard deviation of sampling distribution of proportion is:

Select the correct option

Reload Math Equations

- ○ n/p
- ◉ √(pq/n)
- ○ p
- ○ q/n

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:40 AM, 04 January 2025**

Question # 2 of 10 (Start time: 01:40:39 AM, 04 January 2025) **Total Marks: 1**

If Var(mean1) < Var(mean2) it implies:

Select the correct option

- ○ Mean1 is more efficient
- ○ Mean2 is more consistent
- ○ Mean1 is more consistent
- ◉ Mean2 is more efficient

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:40 AM, 04 January 2025**

Question # 3 of 10 (Start time: 01:41:04 AM, 04 January 2025) **T al Marks:**

A statistic whose standard deviation decreases with an increase in sample size will be called:

Select the correct option

- ○ Sufficient
- ○ Consistent
- ◉ Efficient
- ○ Unbiased

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:40 AM, 04 January 2025**

Question # 4 of 10 (Start time: 01:41:31 AM, 04 January 2025) **T al Marks:**

Which of the following must not be a consistent estimator ?

Select the correct option

- ○ Sample variance
- ◉ Sample median
- ○ Sample Proportion
- ○ Sample mean

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:40 AM, 04 January 2025**

Question # 5 of 10 (Start time: 01:41:55 AM, 04 January 2025) **T al Marks: 1**

When an interval is constructed, what is the probability of occurrence of parameter in it?

Select the correct option

- 1
- 50
- Either 1 or 0
- 0

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:40 AM, 04 January 2025**

Question # 6 of 10 (Start time: 01:42:43 AM, 04 January 2025) **Total Marks:** 1

Which of the following relationship exist between sampling distribution of samples mean and sampling distribution samples median?

Select the correct option

- ○ Both of these
- ○ None of these
- ○ Var(median)< Var(median)
- ◉ Var(mean)<Var(median)

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:40 AM, 04 January 2025**

Question # 7 of 10 (Start time: 01:43:25 AM, 04 January 2025) **T al Marks:**

Statistical Inference can be divided into two main branches:

Select the correct option

- ○ MLE and Methods of moments
- ○ All above
- ◉ Estimation and Hypothesis testing
- ○ Interval Estimation and Point Estimation

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:40 AM, 04 January 2025**

Question # 8 of 10 (Start time: 01:43:50 AM, 04 January 2025) **T al Marks:**

The confidence intervals become wide and less precise by

Select the correct option

- (●) decreasing the sample size
- ( ) decreasing the level of significance
- ( ) increasing the sample size
- ( ) None of these

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:40 AM, 04 January 2025**

Question # 9 of 10 (Start time: 01:44:16 AM, 04 January 2025) **Total Marks:** 1

The mean of the sampling distribution of difference of means is represented by:

Select the correct option

- None of above
- μ
- μ1 -μ 2
- б

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:40 AM, 04 January 2025

Question # 10 of 10 (Start time: 01:44:45 AM, 04 January 2025)

Total Marks: 1

In general, the estimators obtained by the method of moments are .....

Select the correct option

- ○ Inefficient
- ◉ Biased
- ○ All the above
- ○ Inconsistent

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:46 AM, 04 January 2025**

Question # 1 of 10 (Start time: 01:46:36 AM, 04 January 2025) **T al Marks:**

In interval estimation we obtained a .....of values as an estimate of parameter.

Select the correct option

- None of these
- Both a and b
- Group
- Range

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:46 AM, 04 January 2025

Question # 2 of 10 (Start time: 01:47:03 AM, 04 January 2025)

Total Marks: 1

The precision of an estimator can be increased by decreasing the;

Select the correct option

- ○ Number of parameters
- ● Size of sample
- ○ Number of estimators
- ○ Level of significance

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:46 AM, 04 January 2025**

Question # 3 of 10 (Start time: 01:47:26 AM, 04 January 2025) **Total Marks:** 1

The width of the confidence interval is called_____ of the estimate.

Select the correct option

- ○ Unbiasedness
- ○ Biased
- ◉ Precision
- ○ Accuracy

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:46 AM, 04 January 2025**

Question # 4 of 10 (Start time: 01:47:49 AM, 04 January 2025) **T al Marks:**

If Var(mean1) < Var(mean2) it implies:

Select the correct option

- ○ Mean2 is more consistent
- ◉ Mean1 is more efficient
- ○ Mean2 is more efficient
- ○ Mean1 is more consistent

Click to Save Answer & Move to Next Question

BC210411202: WAJID ALI

Time Left 42 sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:46 AM 4 January 2025

Question # 5 of 10 (Start time: 01:48:13 AM, 04 January 2025)

Total Marks: 1

In case of sampling with replacement from two infinite populations, the standard deviation of the sampling distribution of bar X1 - bar X2 standard error of bar X1 - bar X2 denoted by:

Select the correct option

Reload Math Equations

- ○ √((σ1+σ2)/n)
- ◉ √((σ1)2/n1)+ (σ2)2/n2
- ○ √((σ1-σ2)/n)
- ○ 1/n

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:46 AM, 04 January 2025

Question # 6 of 10 (Start time: 01:49:03 AM, 04 January 2025)

Total Marks: 1

Which of the following must not be a consistent estimator ?

Select the correct option

- Sample mean
- Sample Proportion
- Sample variance
- Sample median

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:46 AM, 04 January 2025**

Question # 7 of 10 (Start time: 01:49:31 AM, 04 January 2025) **Total Marks:** 1

The difference between the value of the statistic and the value of the parameter is called _____.

Select the correct option

- ○ Random Error
- ○ Standard Error
- ◉ Sampling Error
- ○ Biased Error

Click to Save Answer & Move to Next Question

BC210411202: WAJID ALI

Time Left 63 sec(s)

STA301 - Statistics and Probability (Quiz 4)

Quiz Start Time: 01:46 AM, 04 January 2025

Question # 8 of 10 (Start time: 01:49:56 AM, 04 January 2025) **Total Marks:** 1

The standard error of the sampling distribution of p1 - p2 is equal to:

Select the correct option

Reload Math Equations

◉	√(p1q1/n1)+ p2q2/n2
○	√((p1+p2)/n)
○	√(p/n)
○	None of the above

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:46 AM, 04 January 2025**

Question # 9 of 10 (Start time: 01:50:27 AM, 04 January 2025) **T al Marks: 1**

From which of the following methods we can obtain point estimate of the population parameters.

Select the correct option

- (●) All of the above
- ( ) Maximum Liklihood Method
- ( ) Method of Least Squares
- ( ) Method of Moments

Click to Save Answer & Move to Next Question

sec(s)

STA301 - Statistics and Probability (Quiz 4) **Quiz Start Time: 01:46 AM, 04 January 2025**

Question # 10 of 10 (Start time: 01:50:53 AM, 04 January 2025) **T al Marks:**

As a rule of thumb, when n>=30, then we can assume that.....is normally distributed:

Select the correct option

- ○ Sampling distribution
- ○ Binomial distribution
- ○ Probability distribution
- ◉ Sampling distribution of sample mean

Click to Save Answer & Move to Next Question